AKSENS Application as Virtual Counseling Encouraging Ideal Educational Characters to Reduce Student Mental Health Decline

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***Abstract*—The mental health condition of students during the COVID-19 pandemic has decreased a lot. Various efforts have been made by counselors to help students reduce academic stress and anxiety, one of which is providing online counseling services. However, online counseling that has been used is still not application-based, especially to prevent and cure mental health setbacks in students. Therefore, this study aims to develop the AKSENS application as a medium for handling student mental health setbacks. The design of this research is the development of the ADDIE model. The research subjects are guidance and counseling experts, IT experts, counselors and students of SMA Negeri 9 Malang. The data collection stage uses questionnaires and interviews with counselors and students. Data analysis used quantitative and qualitative data analysis. The results of the material test are 86.5%, IT experts are 85%, counselor and student user tests are 84.1%, meaning that it can be concluded that the AKSENS application is very suitable to be used as an online counseling service for counselors and students in schools**

Keywords—AKSENS application, online counseling, mental health

I. INTRODUCTION (*HEADING 1*)

A report from ABC news on March 7 , 2020, stated that dozens of countries were closing schools to prevent the spread of the COVID-19 virus. At least 290.5 million students around the world are disrupted in their learning activities because schools are closed, so teachers and students are forced to interact by means of online or online learning. However, the implementation of home study policies makes most students feel anxious and depressed, such as the many assignments given by the teacher in a short time, so that it can make students feel confused and stressed in undergoing online learning [2], [10], [13] , [5]. In addition, there was a case of a high school student in Gowa Regency, South Sulawesi who committed suicide allegedly due to depression due to the large number of online school assignments [4].

The problems experienced by these students can certainly affect students' mental health such as excessive anxiety as well as stress and depression. Thus, the condition of these students is of special concern to the counselor or guidance and counseling teacher to provide interventions to prevent mental health decline. More than 40% of students intend to use online counseling services to help reduce mental health problems experienced by students and the results show that online counseling services are effective and successful in reducing student academic stress and anxiety during the pandemic. COVID-19 [3], 18]. For example, designing, and realizing the model and architecture of e-counseling applications based on social networks to support online student consultation activities [15]. A further effort is to develop an e-counseling application as a guide for the implementation of consultation and treatment for HIV sufferers [14]. There are other studies that have also developed behavioral cognitive cyber counseling applications for guidance and counseling teachers in the industrial revolution 4.0 era from 2006 to 2020 [7].

Researchers created a virtual counseling application with the aim of making it easier for students and teachers to carry out the counseling process. In addition, there is a WFH (Work From Home) policy that limits direct interaction between each other and performs various activities virtually (online). This study focuses on the character of students because it is in accordance with the results of previous and current research which states that the character of high school students (SMA) needs further attention because they are mentally disturbed

Based on the explanation above, researchers are interested in developing an online counseling application that helps prevent students' mental health deterioration. The researchers' interest in developing the application is supported by research data which shows that around 64.9% of respondents choose social media as a mental health service media, and 75.1% of respondents choose other smart device applications in mental health services. This means that most students are interested and enthusiastic about doing online counseling with smart applications to help improve their mental health.

The application that will be developed is a virtual counseling room which aims to tell the problem and consult with counselor or guidance and counseling teachers. This application is devoted to supporting the implementation of

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guidance and counseling services in schools. In this application, there will be a pretest to determine the mental health level of students, discussion forums to share stories and responses anonymously, chat with counselors, and view articles related to mental health given by counselor. With this application, students will get a place to share their feelings, and be able to develop themselves to be even better. Students will find it easier to access virtual counseling rooms because they are more practical and flexible (can be used anytime).

The existence of this application will later help counselor to optimize the ideal character of students in accordance with the Competence Standards for Student Self-Reliance (SKKPD). By optimizing some of these aspects through virtual counseling, it will be a medium for improving mental health in students [11]. The virtual counseling room application will also make it easier for BK teachers to help students optimize the achievement of developmental tasks optimally so that student's mental health is maintained. This study aims to develop a website-based android application designed in a virtual counseling room to encourage students' ideal characters so that they can reduce mental health decline and manual books for AKSENS application users.

II. METHOD

This study uses research and development methods with the ADDIE model [6] including: (1) analysis; (2) designs; (3) development; (4) implementation; (5) evaluation. The flow of the research implementation is shown in Figure 1.

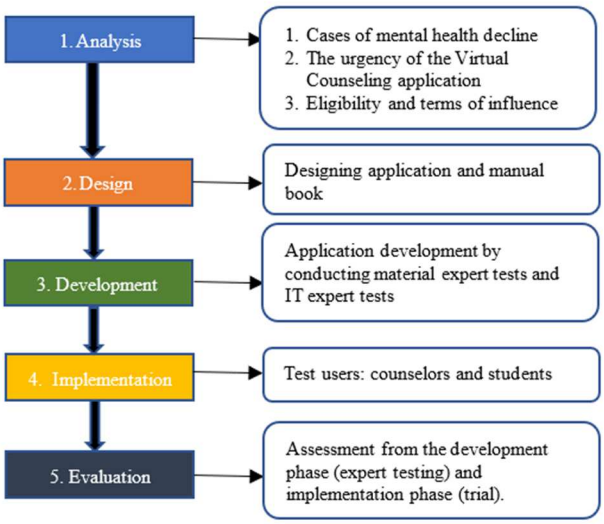

Fig. 1. Research stages

The research subjects involved 2 material experts from guidance and counseling experts, 2 IT experts from web developer development experts, and the trial subject was a counselor and students of SMA Negeri 9 Malang which consisted of 2 counselors and 72 students. SMA Negeri 9 Malang, is one of the state high schools in East Java Province, Indonesia. Application development research uses data collection instruments in the form of questionnaires to obtain assessment data and suggestions from experts and potential product users based on product characteristics of the virtual counseling room research model to encourage students' ideal character in preventing mental health decline. This assessment questionnaire is equipped with a comment and suggestion sheet. Then qualitative data in the form of notes or input from expert experts, and potential product users are used as guidelines for improving the developed product, so that it becomes a product that can be used.

Analysis of the data used quantitatively and qualitatively. The assessment score uses a Likert scale with a score of 1 to 4 and the criteria are very good/very appropriate, good/appropriate, not good/not appropriate, and not good/not appropriate. Then to test the feasibility of the application using the percentage formula as follows [1].

$$Eligibility\ Percentage\ (\%) = \frac{observed\ score}{expected\ score}\ x\ 100 \quad (1)$$

The application eligibility category based on the percentage scale is shown in Table I.

TABLE I. APPLICATION ELIGIBILITY CATEGORY

Percentage (%)	Category
76% - 100%	Very Worthy
51% - 75%	Worthy
26% - 50%	Quite Worthy
0% - 25%	Not Worthy

III. RESULT

A. Expert Validation Results

The AKSENS application was validated by a material expert: Two experts who have many experiences for guidance and counseling; one expert who has experience for IT experts and web developers. The results of material validation by material experts, the AKSENS application got a score of 86.25% which indicates that this application is feasible to be used as a means of implementing counseling in schools. For further development, the validator provides several inputs for the application improvement process:

- The choice of the word decline in mental health is not appropriate, you should use the choice of the word "mental decline".
- The mention of teachers either on the website or in the guide is too general if what is being discussed is mental health. Better specifically called a counselor.
- We recommend that the website also provide features that allow students to contact counselors directly (synchronized live, for example, Google meet/video call).
- It would be even more interesting if the website provided a choice of counselors who could serve students so that students have the choice of a counselor who will help them.
- There needs to be an explanation of the scale of the pretest & post-test this is the result of adaptation/ developing itself. If it is the result of adaptation, then on the website it is necessary to explain who the scale is the result of adaptation.
- It is advisable for the pretest and post-test to use the same instrument because we want to know the difference in conditions before and after receiving counseling services. It should be measured using the same instrument.
- In preparing the answer choices, it is better to use a Likert scale of 4 answer choices, considering that the use of 5 answer choices tends to lead respondents to

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choose the answer choice in the middle, which is sometimes.

- The preparation of pre & post-test items should be arranged using effective sentences and stated in positive sentences. For example "can't" becomes "difficult", the word "do not care" can be replaced with the word "ignorant, or ignore".
- Avoid using the word "feel" too much. It is recommended that the pretest items no 4, 5, 6, 11, 19, 22 and 29 for the word "feel" be omitted. Meanwhile, in post-test items no. 4, 9, 16, 26, 28 and 30, the use of the word "feel" was also omitted.
- In the pretest item no 21, this item needs to be considered because the item statement is difficult for respondents to agree on, because it talks about an increase in body weight of 3 kg in a week.
- On the cover of the guide book for students is written below right "We help teachers to prevent mental health decline" it should be "We help students overcome mental health decline".
- The cover needs an appropriate and more attractive illustration appearance.
- On the second page after logging in there are brief instructions for the activity steps.
- On each page there needs to be a sign to the next activity and return to the previous activity.
- There needs to be an explanation of the purpose and use of forums and articles.
- The word "consult" on the cover of the manual book is not appropriate because consultation is one of the services that is equivalent to counseling. Counseling is assistance services to students that are directly provided to them, while consulting is assistance services to students through parties that influence their development.
- Writing needs to be more careful in terms of language accuracy that is good and correct. For example, a foreign language is italicized, and redirected is concatenated to be redirected.
- On some pages of the accent application there are foreign sentences. The sentence is replaced with an explanation related to the command on the page in question.

The results of the validation of IT experts from the AKSENS application get a score of 85%, meaning that it shows that this application is feasible as a means of implementing counseling services in schools. Further development, the validator asks to fix some bugs and errors that are still encountered:

- Navigation for the chat menu would be better placed on the left sidebar. So, there is no navigation that strays far in the top right topbar.
- For the chat menu on the student account, it would be nice to be able to access it when the pretest has been completed.
- Fixed several bugs such as some devices unable to login to the dashboard page , the feedback feature in the forum displays responses in other forums, if you login using teacher and student accounts simultaneously, the page cannot be found, and in the chat feature the images sent are not uploaded so be an empty file.
- If possible there are some additions to existing features to make the user experience (UX) better. In the chat feature , when sending a picture the name does not need to be displayed. The night mode button is very small, so it is better to combine it with the symbol. Then on the register students tab on the teacher dashboard page , it would be better for the teacher to be able to change the student class table between grades 10 to 12.

B. *Prospective User Test Results*

The AKSENS application is tested for prospective users of counselors and students. The test results for prospective counselor users were carried out by two counselors and 72 students with a score of 84.1%. The inputs obtained from the BK teacher/counselor are: If possible, personal counseling services can be added, scheduling counseling services, videos as apperception, general knowledge about counseling and the principles in it. If possible, a behavior contract can be added. Sometimes the page still has errors so it needs to be fixed again. Sometimes can't login. While the input obtained from students in a nutshell:

- Websites sometimes still have errors and need a strong network. But sometimes also can't login at all.
- Page loading is quite long.
- Images in the chat room can be seen immediately, and if possible, it is necessary to add a reply chat and voice note feature.
- The color of the website display needs to be more colorful.
- The choice of profile photos is made more diverse.

C. *AKSENS Virtual Counseling Website Application Development*

The design of an android website application based on a virtual counseling website called AKSENS (Aplikasi Kesehatan Mental Siswa) which has its own logo as shown in Figure 2.

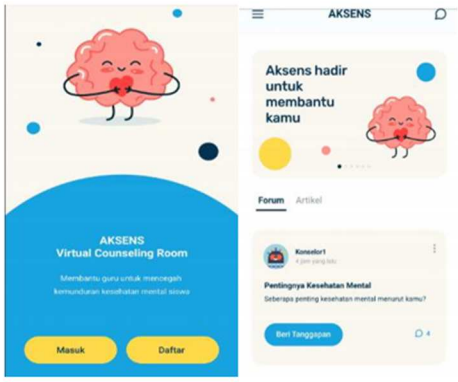

Fig. 2. AKSENS display picture

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After designing the display, the AKSENS logo was also developed and the appearance looks like in Figure 3.

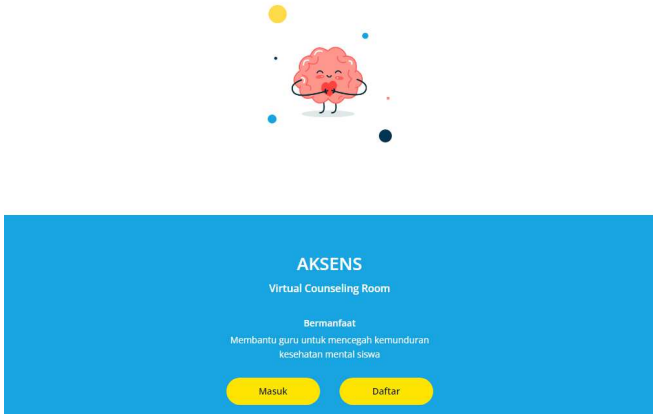

Fig. 3. AKSENS logo development

This is the initial display for counselor when opening the AKSENS website link for teacher users , namely https://aksens.herokuapp.com/gr . Next, counselor needs to log in, and if you don't have an account yet, you need to select the “register” button to create an account first

Meanwhile, this is the initial display for students when they open the website link accents.herokuapp.com . Students can login using an account that has been created by the counselor. For further information, as shown in Figure 4.

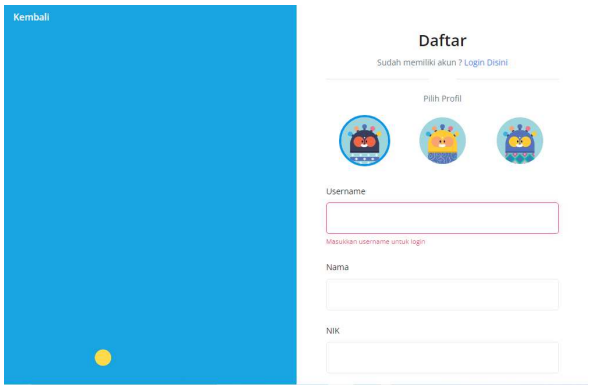

Fig. 4. List display for the counselor user

The picture above is a list display for the counselor user. For the login options for the teacher and student users it looks exactly like the picture. It only differs in the input form only. The picture is a display of the counselor dashboard. The student dashboard is also exactly the same except for the side bar menu , where for student users there are only options for dashboard , profile, pretest , post-test , and create a forum. For further information, as shown Figure 5.

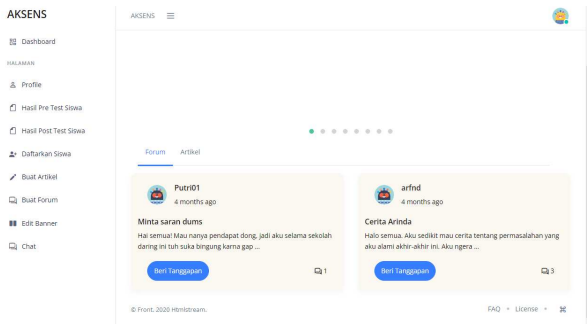

Fig. 5. Display of the counselor dashboard

After logging in and filling out the profile, the counselor can register students by selecting the “Register Students” button in the side bar. In contrast to the counselor user. After logging in, the student user will be directed to fill out a new pretest then can complete the profile and be directed to the dashboard . The following is a display of the pretest and post-test menus from student users. More clearly contained in the following Figure 6.

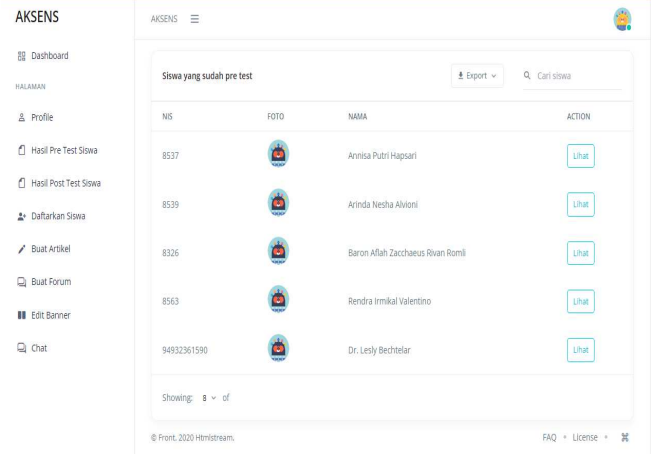

Fig. 6. Display of the pretest and post-test menus from student users

After filling out the pretest or post-test menu , students can see the results as shown above. From the pretest and post-test, students will be given a brief explanation regarding their mental health condition. More clearly contained in the following figure 7.

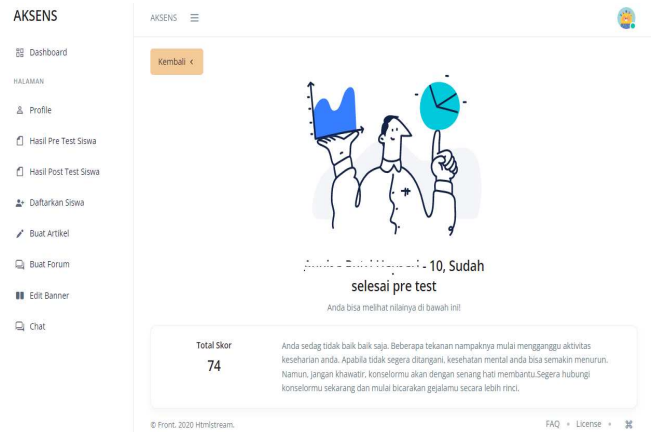

Fig. 7. Explanation regarding their mental health condition

The development that was made was not only the AKSENS application, but also a manual book for using the application for counselor and student users. The following are the results of the design of the AKSENS application guidebook as shown in figure 8.

The following is a display of the chat features that can be accessed by counselor and student users. In addition to the chat feature, there is also a "View Articles" display from student users. This article can be added and edited directly by the counselor. For further information, as shown in Figure 9. Students can create discussion forums related to problems that are being experienced or respond to forums that have been created by other students as shown above. In this forum feature, counselor users can also access and create forums or respond.

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Fig. 8. AKSENS application guidebook

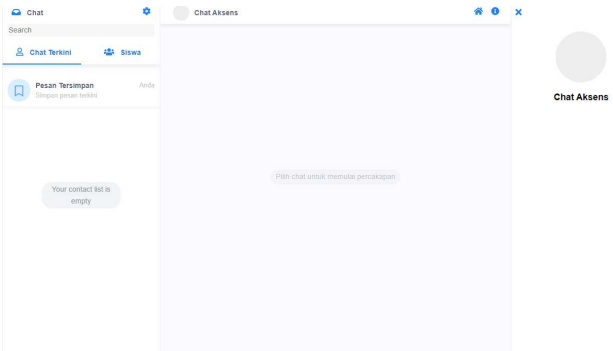

Fig. 9. Display chat features and articles

The manual book consists of: (1) cover , showing the initial appearance of the manual book consisting of the application title, logo, and supporting images; (2) Table of Contents, there are titles of the sections of the application guidebook; (3) Introduction, there are background, scope, benefits, and use of the application; (4) Instructions for using the AKSENS website application for teachers, there are stages of using the application for teachers accompanied by pictures (5) Instructions for using the AKSENS website application for students, there are stages of use applications for students accompanied by pictures; and (6) Closing, there is hope for application users as well as suggestions and feedback that support this application. The manual is saved in soft file or pdf format

IV. Discussion

AKSENS is a virtual counseling website application that is devoted to assisting school counselor (guidance and counseling teachers) in conducting the counseling process with students. This application is devoted to supporting the implementation of guidance and counseling services in schools. With this students will get a place to share their feelings, and be able to develop themselves to be even better. Students will find it easier to access virtual counseling rooms because they are more practical and flexible (can be used anytime)

AKSENS consists of software (software and system) and hardware (equipment). Software includes virtual counseling websites and applications , and softcopy manuals. As for the hardware, it consists of the AKSENS website application guide book. The implementation of this online service is very possible for counselors to provide 24-hour services to students and also parents of students, which is not possible to come to consult directly with counselees face to face , counselors can perform services anywhere without having to stand by in the workspace [8].

Counselors have hours, days, or even weeks to respond to the message. Opportunities to send messages to counselors can be done at any time. In line with the results of research by Dincyurek & Gulen, academics view that online counseling services will be useful for shy students who cannot come to school counseling services [9]. In addition, the online guidance and counseling service can provide 24 hours a day. This will provide an opportunity to reach the counselee more broadly. And the counselor can provide direct services to the counselee together. The use of AKSENS also has a positive effect where students can communicate regularly with counselors, wherever and whenever. In addition, with the features in AKSENS, it certainly makes the reach of counseling easier.

Based on the results of the study, it shows that guidance and counseling services can also attract the interest of many people, especially the millennial generation. Counselors must be able to hone their skills even more deeply to take advantage of technology and create new innovations. With this, guidance and counseling services will continue to develop and provide benefits for many people. The advantage of counselors in providing counseling services with the help of technology is that counselors have the ability to use other technology sources that can be used in counseling guidance [19]. This can improve the ability of the counselor himself and himself and build prestige among other teachers in the school.

The development of virtual guidance and counseling services can increase student interest in participating in guidance and counseling services. In addition, the millennial generation who often uses information technology will also know and understand more about guidance and counseling services. With this, guidance and counseling services can be adaptive, that is, they can adapt to the needs of students in this era of disruption. Several developments in guidance and counseling services are efforts made to be able to answer the world's needs as a result of the industrial revolution 4.0, technology in guidance and counseling services has a very large contribution to the implementation of BK services, the role of technology in guidance and counseling includes making it easier to design, process, and process data related to the services [12]. The application is used because it can help students improve their learning because it can be accessed whenever and wherever students are [20].

V. Conclusion

Based on the results of the study, it can be concluded that the AKSENS application has been running quite optimally and effectively. For counselor users and students, they are helped when conducting online counseling, especially during the COVID-19 pandemic. The results of the material expert test, IT expert and the test of counselor and counselee users are on average in the range of 76%-100% which means it is very good and this is considered very feasible applied to help reduce the mental health decline of SMA Negeri 9 Malang students.

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